Homework 4
Introduction to Complex Analysis
(MATH 4230-40), Spring 2022

Section 1.6 - Differentiation of the Elementary Functions

1. Determine whether the following complex limits exist and find their value if they do:
   (a) $\lim_{z\to 0} \frac{e^z - 1}{z}$
   (b) $\lim_{z\to 0} \frac{\sin|z|}{z}$

2. Let $u(x,y)$ and $v(x,y)$ be real valued functions on an open set $A \subseteq \mathbb{R}^2$ and suppose they satisfy the Cauchy-Riemann equations on A. Show that this implies that the following pairs of functions satisfy the Cauchy-Riemann equations on A as well:
   (a) $u_1 := \big(u(x,y)\big)^2 - \big(v(x,y)\big)^2$ and $v_1(x,y) := 2u(x,y)v(x,y)$
   (b) $u_2(x,y) := e^{u(x,y)}\cos\big(v(x,y)\big)$ and $v_2(x,y) := e^{u(x,y)}\sin\big(v(x,y)\big)$

3. Find the region of analyticity and the derivative of each of the following functions:
   (a) $\frac{z}{z^2-1}$
   (b) $e^{z+(1/z)}$